Supplementary

1 Progression Attributes and Residual Mask Extraction

Progression Attributes: The inputs to the condition encoder $\mathcal{C}$ are the progression attributes comprising of - (i) the age gap (v_a) between the previous time-point and follow-up, and (ii) the cognitive status (v_d). These attributes are encoded as one-hot vectors given as,

$$({v_d}^i, {v_a}^j) = ((0, \ldots, 1_i, \ldots, 0), (0, \ldots, 1_j, \ldots, 0)) \tag{1}$$

where i and j denote the indices corresponding to the cognitive subtype and age-gap bin, respectively. The cognitive categories include CN, MCI, and AD, while age gaps are discretized in 0.5-year intervals. The maximum age gap is determined by the availability of longitudinal samples in the training dataset.

Residual Mask Extraction: The residual mask (M) is obtained by computing the residual between the baseline and ground-truth follow-up images, followed by thresholding. The threshold is selected to capture ventricular expansion (mostly observed in the residuals of this dataset), which appears as hypo-intense regions in the residual maps, consistent with forward-time disease progression and the axial modeling direction. Specifically, the residual is normalized to the range $[0, 1]$, and a threshold of 0.15 is applied, which was empirically observed to cover most ventricular expansion regions in this dataset. While this choice is dataset-specific, it provides a reasonable approximation and can be made more robust in future extensions of this work.

2 Parameter Details

Table 1: Hyperparameters and configuration details used in our experiments.

Parameters	**Values**	**Parameters**	**Values**
Axial Height (H), Width (W), Slices (D)	208, 160, 100	Inference Diffusion Steps (T_s)	50
Latent Dimension (d)	512	Decoder Layers for Alignment (L)	3
Conditioning Vector Dimension (d')	50	Decoder Layer Dimensions (d_k, h, w)	512, 16, 16
Training Diffusion Steps (T)	1000	Loss Weights ($\lambda_1, \lambda_2, \lambda_3$)	0.01, 0.01, 1
Epochs	180	Optimizer	Adam
Learning Rate	0.001	Diffusion Noise Schedule	Linear

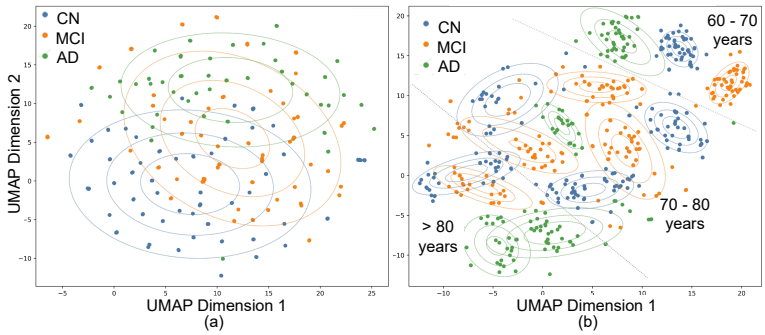

Figure 1: The 2D UMAP projections of different latent subspaces of our model are shown from left to right: (a) the identity-preserving latent subspace $((z_b + z^{'})[d^{'} : d]))$, and (b) the progression-related latent subspace $(z_b + z^{'})[0 : d^{'}])$. The projection in the sub figure (b) exhibits clear alignment with cognitive status and age groups, indicating that progression-related information is encoded within $(z_b + z^{'})[0 : d^{'}])$.

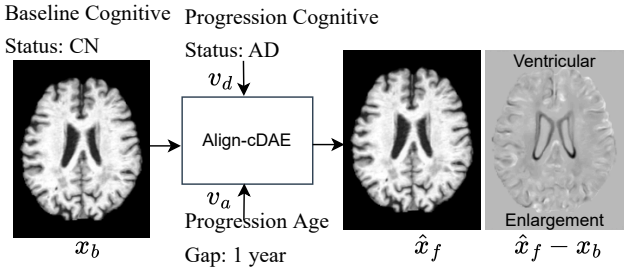

Figure 2: Left to right: (i) input image from the CN category, (ii) progression attributes corresponding to the AD category with a 1-year age gap, (iii) generated follow-up image, and (iv) residual map highlighting significant progression-related changes (ventricular expansion)

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3 Latent Space-based Ablation Study

The latent subspaces of our model, which separately encode disease-progression–related information and subject-identity–related details, are projected into a two-dimensional space and shown in Figure 1. Sub-figure (a) presents the identity-preserving latent subspace $((z_b + z^{'})[d^{'} : d]))$, while sub-figure (b) shows the progression-related latent subspace $(z_b + z^{'})[0 : d^{'}])$. From sub-figure (b), distinct clusters aligned with cognitive status and age groups are observed, indicating that progression-related information is organized within $(z_b + z^{'})[0 : d^{'}])$. Such a

structure is not clearly delineable in sub-figure (a). Additionally, sub-figure (b) shows stronger separability among cognitive statuses at lower age ranges, whereas at higher ages the clusters exhibit increased overlap, reflecting the reduced separability between normal aging and disease-related progression.